MM7025 - COMPUTER ALGEBRA, COMPUTER EXERCISES 1

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Please work in pairs or in triples.

(1) Implement a function in Macaulay2 that returns the n'th Fibonacci number. Note: Speed is not of the essence.

(2) (a) Compute first by hand, and then by Macaulay2 using the "%"-operator, the remainder when $x^7y^2 + x^3y^2 - y + 1$ is divided by $xy^2 - x$ using lex (M2: Lex), glex (M2: GLex), and grevlex (M2: GRevLex, the standard ordering), all with respect to $x > y$.

(b) Compute first by hand, and then by Macaulay2 using the "%"-operator, the remainder when $x^7y^2 + x^3y^2 - y + 1$ is divided by $x - y^3$ using the three above monomial orderings, all with respect to $x > y$.

(c) Compute first by hand, and then by Macaulay2 using the "%"-operator, the remainder when $x^7y^2 + x^3y^2 - y + 1$ is divided by $\{x - y^3, xy^2 - x\}$ using the three above monomial orderings, all with respect to $x > y$. This means that you should continue applying the "%"-operator with one divisor at a time, until you get a remainder that is divisible by neither $\mathrm{LM}(x - y^3)$ nor $\mathrm{LM}(xy^2 - x)$.

(3) Implement a function in Macaulay2 that takes as input a prime p and a positive integer d and returns the number of distinct irreducible factors of the polynomial $x^d - 1$ in $\mathbb{Z}_p[x]$. Hint: To define the polynomial ring over $\mathbb{Z}_p$, use `ZZ/p[x]`.

For help on how to apply the factor function, see `https://macaulay2.com/doc/Macaulay2/share/doc/Macaulay2/Macaulay2Doc/html/_factoring_sppolynomials.html` Optional: Experiment with the function and see if you can find some pattern.

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